

Module 3 – Machine Learning

1. Machine Learning Introduction

What is Machine Learning?

Machine Learning is a method where computers learn from data and make predictions or decisions without being explicitly programmed.

Why It Matters

- Automates decision-making
 - Finds patterns in data
 - Powers modern AI systems
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2. Supervised Learning

What is Supervised Learning?

A type of machine learning where models are trained using labeled data.

Examples

- Predicting house prices
- Email spam detection

Key Idea

Input data + correct output → model learns mapping

3. Unsupervised Learning

What is Unsupervised Learning?

A type of learning where the model works with unlabeled data.

Examples

- Customer segmentation
- Pattern detection

Key Idea

Model finds hidden structure in data

4. Model Training

What is Model Training?

Training is the process of feeding data into a model so it can learn patterns.

Steps

- Provide dataset
 - Train model
 - Adjust parameters
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5. Evaluation Metrics

What are Evaluation Metrics?

Metrics are used to measure how well a model performs.

Common Metrics

- Accuracy
 - Precision
 - Recall
 - F1 Score
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6. Key Concepts to Remember

- Machine learning learns from data
 - Supervised learning uses labeled data
 - Unsupervised learning uses unlabeled data
 - Training helps model learn patterns
 - Metrics evaluate performance
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7. Summary

Machine learning is the core of data science. By understanding supervised and unsupervised learning, model training, and evaluation metrics, you can build intelligent systems that make accurate predictions.